Basic Angle Properties

Exam code: 0607

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Written by: Jamie Wood
Reviewed by: Dan Finlay
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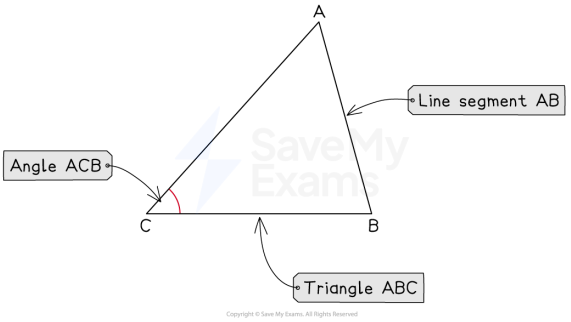
Basic angle properties

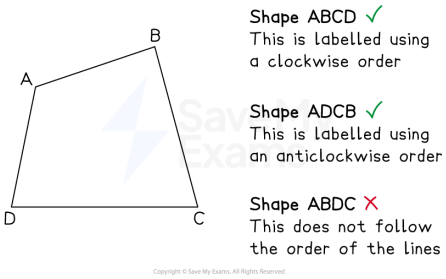
How do I label lines segments, angles and shapes?

- The **line segment** *AB* goes from point *A* to point *B*
- The **angle** *ABC* is formed by the line segments *AB* and *BC*
  - The angle is at point *B*
  - The angle is the acute or obtuse angle, not the reflex angle
- The **triangle** *ABC* is formed by the line segments *AB*, *BC* and *CA*



Labels for line segments, angles and triangles

- The **quadrilateral** *ABCD* is formed by the line segments *AB*, *BC*, *CD* and *DA*
  - The points are labelled starting at a point and following the sides
  - *ABDC* is **not** the same shape because it is formed by the line segments *AB*, *BD*, *DC* and *CA*
- The same idea is used to label other **polygons**

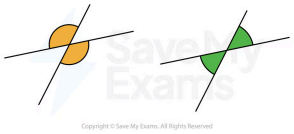


Labels for shapes

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

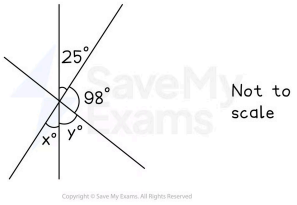
- Angles around a **point** add up to **360°**
- Angles that form a **straight line** add up to **180°**
- **Vertically opposite** angles are **equal**
  - Vertically opposite angles occur when **two lines intersect**, as in the diagram below



Vertically opposite angles

Worked Example

The diagram below shows three straight lines intersecting at a point.



Find the values of x and y.

Answer:

Vertically opposite angles between two intersecting lines are equal

$$x = 25$$

Angles that meet on a straight line add up to 180°

$$x + y + 98 = 180$$
$$25 + y + 98 = 180$$
$$123 + y = 180$$

Solve to find y

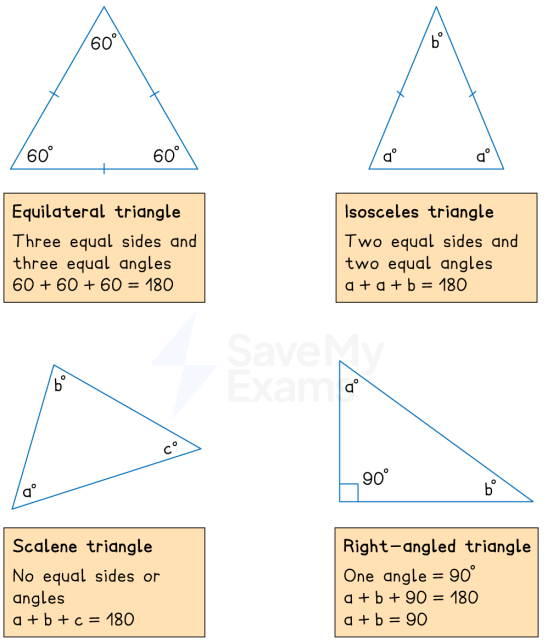
$$y = 180 - 123$$
$$y = 57$$

$$x = 25 \text{ , } y = 57$$

What are the angle properties of triangles?

- The **three** interior angles inside any triangle **add up to 180°**

- If the triangle is **equilateral** then all three angles will be equal
  - Each angle will equal **60°**
- A **right-angled triangle** has one 90° angle



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Examiner Tips and Tricks

Find all the missing angles that you can using the angles that are given to you in a question

- They might not seem to help you straight away but having more angles will lead you to find the angle you need

Worked Example

The diagram below is formed using three straight lines. Find the value of x.

Answer:

Vertically opposite angles between two intersecting lines are equal

$$y = 60$$

Angles that meet on a straight line add up to 180°

$$z + 130 = 180$$
$$z = 50$$

Interior angles in a triangle add up to 180°

$$x + 60 + 50 = 180$$
$$x + 110 = 180$$

$$x = 70$$

What are the angle properties of quadrilaterals?

- The **four** interior angles inside any quadrilateral **add up to 360°**
- If the quadrilateral is a **square** or a **rectangle** then all the angles are equal to **90°**
- You can use any **symmetries** of the quadrilateral to identify other equal angles
  - For a **parallelogram** (or rhombus), **opposite angles** are **equal**
  - For a **kite, one pair** of **opposite angles** are **equal**

Worked Example

The diagram below shows an irregular quadrilateral. Find the value of y.

Answer:

First, add together the three given angles

$$97 + 115 + 85 = 297$$

Subtract the answer from 360°

$$360 - 297 = 63$$

Add this to the diagram

$$y + 63 = 180$$
$$y = 180 - 63$$
$$y = 117$$

$$y = 117$$

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